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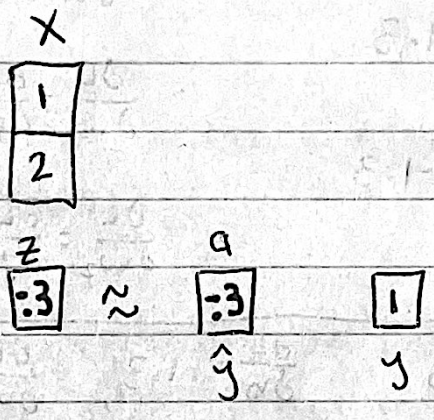
Machine Learning in Medicine and Biology

Homework 4 - Deep Learning for Fun and Profit

Part A - Backpropagation

A.3



1) $z = w_1 * x_1 + w_2 * x_2 + b$

$= .1(1) + (-.2)(2) + 0 = .10 - .40 = -.30$

$\boxed{z = -0.3}$

→ generally, a or $\hat{y}$ would be z after ReLU (which would be 0)

However,

$\hat{y} = z = -.30$

$\boxed{\hat{y} = -0.3}$

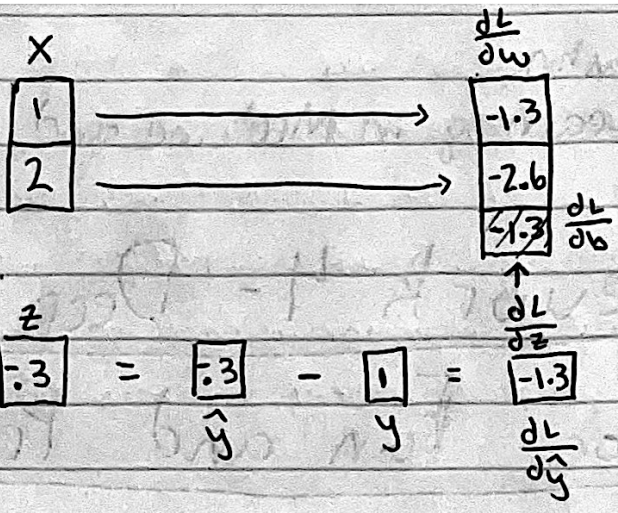
2) $L = \frac{1}{2}(\hat{y} - y)^2$

$= \frac{1}{2}(-.30 - 1.0)^2 = \frac{1}{2}(-1.3)^2 = \frac{1}{2}(1.69) = .845$

$\boxed{L = 0.845}$

(cont.) →

3)



$\frac{dL}{d\hat{y}} = \hat{y} - y = -.3 - 1.0 = -1.3$

$z = \hat{y}, \quad \frac{dL}{dz} = \frac{dL}{d\hat{y}} = -1.3$

$\frac{dL}{db} = \frac{dL}{dz} * 1 = -1.3 \times 1 = -1.3$

$\frac{dL}{dw} = \frac{dL}{dz} * X$

$\frac{dL}{dw_1} = \frac{dL}{dz} \times X_1 = -1.3 \times 1 = -1.3$

$\frac{dL}{dw_2} = \frac{dL}{dz} \times X_2 = -1.3 \times 2 = -2.6$

ε: ε:

$$\frac{dL}{d\hat{y}} = -1.3, \quad \frac{dL}{dz} = -1.3, \quad \frac{dL}{db} = -1.3, \quad \frac{dL}{dw_1} = -1.3, \quad \frac{dL}{dw_2} = -2.6$$

4) $w_{1\,new} = w_1 - \eta\left(\frac{dL}{dw_1}\right) = .1 - .1(-1.3) = .1 + .13 = .23$

$w_{2\,new} = w_2 - \eta\left(\frac{dL}{dw_2}\right) = -.2 - .1(-2.6) = -.2 + .26 = 0.06$

$b_{new} = b - \eta\left(\frac{dL}{db}\right) = 0 - .1(-1.3) = 0 + .13 = 0.13$

$$w_{1\,new} = 0.23, \quad w_{2\,new} = 0.06, \quad b_{new} = 0.13$$

Part B - Colab Notebook

1) Step 1 is to clear the state of the (recorded) gradients between the batches that the data was split in to. This starts the calculations of each batch as their own, rather than a continuation of the previous batch's calculations.

Step 2 is getting the predicted y (or output) from the model.

Step 3 calculates the loss, or the difference between the predicted y and the actual y.

Step 4 performs back propagation on the model. This uses the difference between predicted y and actual y (from Step 3) to get $\frac{dL}{dz}$, ($\frac{dL}{da}$,) $\frac{dL}{db}$, and $\frac{dL}{dw}$ for each level of the model.

Step 5 gets the new weights (and biases) based on Step 4's calculations. This can be represented by $w_{new} = w - \text{learning rate}(\frac{dL}{dw})$ (and $b_{new} = b - \text{learning rate}(\frac{dL}{db})$).

These steps can induce "learning" by causing the model to update and become iteratively better/more accurate. These five steps basically complete the backpropagation by hand steps in part A (reset gradients, forward pass, loss, back-propagation, and gradient descent updates) and allow the model to "learn" and change from its mistakes.

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2) Residual connection in a neural net is the re-introduction of the original input or output of a (more) distant hidden layer before the neural net finalizes the output. This allows the output to be more closely related/influenced by (more) unmodified data. You might want a residual connection between layers because it could produce a more accurate and grounded final output.

3) The label had to be converted from a string to an integer to a tensor because it allowed the label to be moved to the GPU.

4) The observation type they both address CO (carbon monoxide) level in terms of low (< 2.0), medium (> 2.0 and < 5.0) and high (> 5.0). The 1D CNN and RNN take in the same data and both predict [redacted] CO(GT) amount class for a subsequent hour based on the previous 24-hour window. One model is better than the other when its validation accuracy (correct/total) is higher. The 1D CNN is better when number of epochs is 2, 5, 6, 7, 8, and 10. RNN is better when number of epochs is 1, 3, [redacted] 4, and 9. (This could be (very) generalized to RNN is better when there are 4 or fewer epochs, and 1D CNN is better when there are 5 or more epochs.)